Nandan Bharatkumar Parikh

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Education

Georgia Institute of Technology, Master of Science in Computer Science — August 2025 – May 2027

- **Coursework**: Computer Vision, Machine Learning, Efficient ML
- **Research**: Working with Prof. May Wang on creating explainable vision models using concept bottlenecks

Birla Institute of Technology and Science, Pilani, B.E in Computer Science — August 2019 - May 2023

- GPA: **8.8**/10, Major GPA: **9.48**/10
- **Thesis**: High Throughput Microscopy Image Deblurring
- **Coursework**: Deep Learning, Operating Systems, Applied Statistical Methods, Image Processing

Experience

Software Engineer II (SDE-2), **Flipkart** (Hotels Search) – Bengaluru, India — July 2023 – July 2025

- Built an **LLM-powered hotel search system** using **RAG with Gemini API**, integrated via **LangChain4j and Elasticsearch vector DB**, deployed on **GCP (Docker/Kubernetes)** for natural language queries.
- Scaled **Java/Spring** services for **5x traffic** (5k → 24k RPS) during sale events by identifying bottlenecks through load testing, tuning threadpool configurations, and implementing strategic caching layers.
- **Search system revamp**: Optimized the search result page p95 latency by **50% (2s to 900ms)** through **Redis/Aerospike caching** and efficient asynchronous cache refresh with **Kafka/PubSub**.
- Reduced **zero search results by 35%** by building a click-tracking pipeline for autosuggest optimization and enabling support for informally-defined geographic areas.
- Promoted early from SDE-I for outstanding contributions in search systems and AI integration.

Undergraduate Research Assistant, Harvard – Boston, MA — Nov 2022 – June 2023

- Created an **end-to-end microscopy image deblurring** pipeline including **blur detection and correction** to increase microscope throughput using **OpenCV, PyTorch, and CUDA**
- Implemented a novel **graph reasoning attention network (GRAN)** for improved semantic representation
- Achieved **20x inference speedup** over SOTA methods while maintaining competitive image quality (PSNR: 33.23 dB, SSIM: 0.8837).
- **Published** at *Microscopy and Microanalysis*

Software Engineering Intern, Flipkart – Bengaluru, India — Jun 2022 – Aug 2022

- Built a **Python wrapper** around the **MobSF** framework and integrated it into Flipkart's internal security tools.
- Developed a self-service portal using **Google Apps Script, HTML, and CSS** to automate Drive security scans.
- Created a **Python CLI tool** with **Google APIs** to detect external file sharing, cutting manual audits by **60%**.

Publications

- R. L. Schalek, **N. Parikh**, Y. Wu, J. W. Lichtman, D. Wei, *Real-time Image Deblurring to Improve Throughput of Serial-Section Volume Electron Microscopy*, Microscopy & Microanalysis 2023. DOI

Projects

SONAR to Satellite Image Translation — GitHub

- Developed **Pix2Pix image translation model** with custom **enhancement module** and **edge-guided loss**
- Implemented **multiscale discriminator** to reduce artifacts and improve translation quality
- Improved **FID by 1.1%** (71.584 → 70.815) and **PSNR by 3.4%** (31.76 → 32.85)

Technologies

Languages: Python, SQL, Java, MATLAB, C/C++, Git

Technologies: ElasticSearch, Redis, MongoDB, Vertex AI, GCP, Aerospike

Libraries: PyTorch, Tensorflow, Numpy, Pandas, OpenCV, Scikit-learn, Matplotlib, LangChain